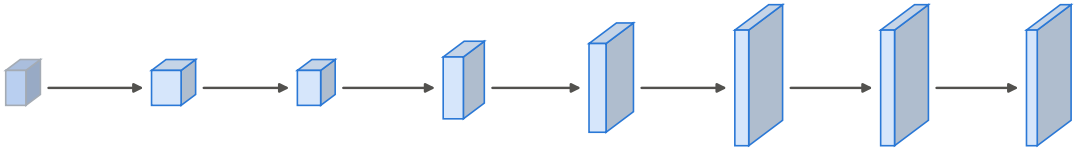
(a) Segmentation decoder: tensor shapes (channels × depth × height × width)



	block tokens (4 taps)	1×1×1 conv, concatenate	3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	upsample, 3^3 conv	1×1×1 conv
TAP-CT	$768\times3\times28^2$	$4\cdot128\times3\times28^2$	$128\times3\times28^2$	$64\times6\times56^2$	$32\times12\times112^2$	$16\times12\times224^2$	–	$C\times12\times224^2$
SPECTRE	1080×8^3	$4\cdot128\times8^3$	128×8^3	64×16^3	32×32^3	16×64^3	$16\times64\times128^2$	$C\times64\times128^2$
VoCo	192×24^3… 1536×3^3	$4\cdot128\times24^3$	128×24^3	64×48^3	32×96^3	–	–	$C\times96^3$

Taps: blocks 3, 6, 9, 12 (TAP-CT) or 6, 12, 18, 24 (SPECTRE) after the final norm; for VoCo the four stage outputs 192×24^3, 384×12^3, 768×6^3 and 1536×3^3 (strides 4 to 32, channel-normalised), resized to the stride-4 grid before concatenation. Each 3^3 conv is followed by instance norm and LeakyReLU; trilinear upsampling along the axes whose stride is still above 1; no skips from the input. SPECTRE: per-crop ViT-L only. Decoder parameters: 2.45 / 2.62 / 2.42 M (TAP-CT / SPECTRE / VoCo, C = 15). Lin: a 1×1×1 conv on the deepest tap (block 12, block 24, or the VoCo stage-4 output at stride 32, i.e. 48 mm) with trilinear upsampling; 769 / 1081 / 1537 parameters per structure. C = 1 per structure, or the number of structures (joint).

(b) What each segmentation method trains

			early blocks			late blocks				encoder parameters updated		
		patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE	VoCo
Lin									1×1×1 conv	8000	0	0	0
Dec									decoder	8000	0	0	0
LoRA									decoder	8000	0.59 M	2.1 M*	0.18 M
LoRA$_{late}$									decoder	8000	0.20 M	0.69 M	0.10 M
DoRA	(+ m)		m			m			decoder	8000	0.62 M	2.2 M*	0.19 M
Dec→LoRA			2			2			decoder	4000 + 4000	0.59 M	2.1 M*	0.18 M
Full FT									decoder	8000	87 M	339 M*	74 M

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

AMOS22 and TotalSegmentator joint models: 24 000 steps (12 000 + 12 000). *SPECTRE: LoRA, DoRA and Full FT also wrap or unfreeze the feature combiner, which never enters the segmentation features and receives no gradient; counting it, 2.4 M / 2.5 M / 397 M.